# What Actually Helps a Training-Free Diffusion Predictor? A Falsification Study of Token-Adaptive Prediction on an Exact-Residual Testbed

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**Abstract.** Training-free diffusion accelerators such as the Token-Adaptive Predictor (TAP) reconstruct skipped network outputs from cached features, choosing *which* cheap predictor to trust, per token, by reading a first-layer *probe* and ranking predictors with a cosine proxy. We set out to improve three levers that looked promising in a single-seed pilot—an $L_2$ proxy in place of cosine, online least-squares predictors in place of fixed Taylor, and the probe's per-token adaptivity—and instead *falsify* most of them. On an **exact-residual Gaussian-mixture testbed**, where the true denoiser output is closed-form (Tweedie) so proxy fidelity is directly measurable without a trained network, we run a 12-seed core study, a 21-regime sweep, and a pre-registered 30-test adversarial suite (all with bootstrap 95% CIs). Reported straight: **(i)** the $L_2$ proxy beats cosine in rank fidelity in *all* 21 regimes, but the effect is small ($\Delta$ Spearman $+0.016$, CI $(+0.014, +0.019)$; not the $\sim 4\times$ of the pilot) and grows only as the task hardens; **(ii)** online predictors do *not* beat fixed Taylor—they are reliably worse ($19/21$ regimes); **(iii)** probe-then-select beats a strong fixed predictor *only* at aggressive acceleration, so at moderate speedups the speedup—not the adaptivity—does the work. We prove that per-token selection is already optimal (separable, equal-cost, unconstrained $\Rightarrow$ greedy $=$ optimum), reframing the real object of study as *proxy fidelity*: an oracle selector leaves headroom the probe cannot capture, and fidelity rises monotonically with probe richness ($0.33 \to 1.0$). We give the mechanism for why an $L_2$ proxy tracks the $L_2$ output residual better than a magnitude-blind cosine.

*TAP (arXiv:2603.03792) has no public code; all TAP behaviour here rests on our faithful reimplementation. The testbed is a shallow analytic denoiser and deliberately lacks the deep learned layers that make a real first-layer probe more informative; we quantify that gap rather than hide it. All code and numbers are released.*

## 1 Introduction

Feature-caching accelerators for diffusion transformers (DiTs) [5] skip most of the network on most sampling steps and reconstruct the missing output from cached values [8, 9]. The Token-Adaptive Predictor (TAP) adds a twist: it keeps a *family* of cheap extrapolators and, per token, selects one by reading the first-layer *probe* (the modulated input) and ranking predictors by how well their extrapolated probe matches the freshly computed probe, under a cosine proxy. A single-seed pilot of ours suggested three ways to do better: swap cosine for $L_2$, replace fixed Taylor extrapolators with online least-squares, and lean on the probe's adaptivity. This paper is the adversarial follow-up. Rather than advertise the levers, we try to *break* them under multi-seed, multi-regime pressure, and report what survives.

The result is deliberately unglamorous and, we argue, more useful than another cache heuristic. Two of the three pilot levers do not survive contact with multiple seeds; the third survives only as a small, robust edge. The study also clarifies *what the real degree of freedom is*: because TAP's per-token selection is separable, equal-cost, and unconstrained, greedy per-token selection is already the global optimum (§4); there is no selection-optimization gap to close. The binding constraint is the fidelity of the cheap proxy to the true, unobserved output error—a quantity our exact testbed lets us measure directly.

**Contributions.**

- An *exact-residual* Gaussian-mixture diffusion testbed (Fig. 1) in which the true denoiser output, and therefore every predictor's true error, is closed-form, making proxy fidelity measurable without a trained network.
- A proof that TAP's per-token selection has *zero* optimality gap, relocating the problem from selection optimization to proxy fidelity, plus a mechanism for why an $L_2$ proxy dominates cosine.
- A rigorous falsification: 12-seed core, 21-regime sweep, and a pre-registered 30-test suite (Fig. 2) that retire the online-predictor and probe-adaptivity claims and bound the $L_2$ claim to a small, robust edge.

## 2 Background and related work

**Feature caching.** DeepCache [8] and FORA [9] reuse features on fixed schedules; TeaCache [10] modulates reuse by a timestep-embedding indicator; ToCa [11] and DiCache [12] make the decision token- or sample-adaptive; TaylorSeer [13] replaces cache-and-reuse with Taylor *prediction* of features. TAP generalises the predictor to a per-token *selected* member of a small family. **Diffusion samplers.** We use the variance-preserving forward process of DDPM [1] with deterministic DDIM [2] reverse steps; the exact posterior mean follows from Tweedie's formula [6] and the score view of diffusion [3, 4]. **Evaluation hazard.** These methods are compared by generation quality at matched compute; a recurring hazard, which we make explicit and which motivated the present study, is that comparisons on analytic or untrained denoisers need not transfer to trained ones—a companion study shows the probe's reliability is itself an emergent property of training [14]. **Status of TAP.** TAP (arXiv:2603.03792) has no released implementation at the time of writing; every TAP-specific claim here rests on our faithful reconstruction of its probe-then-select mechanism,

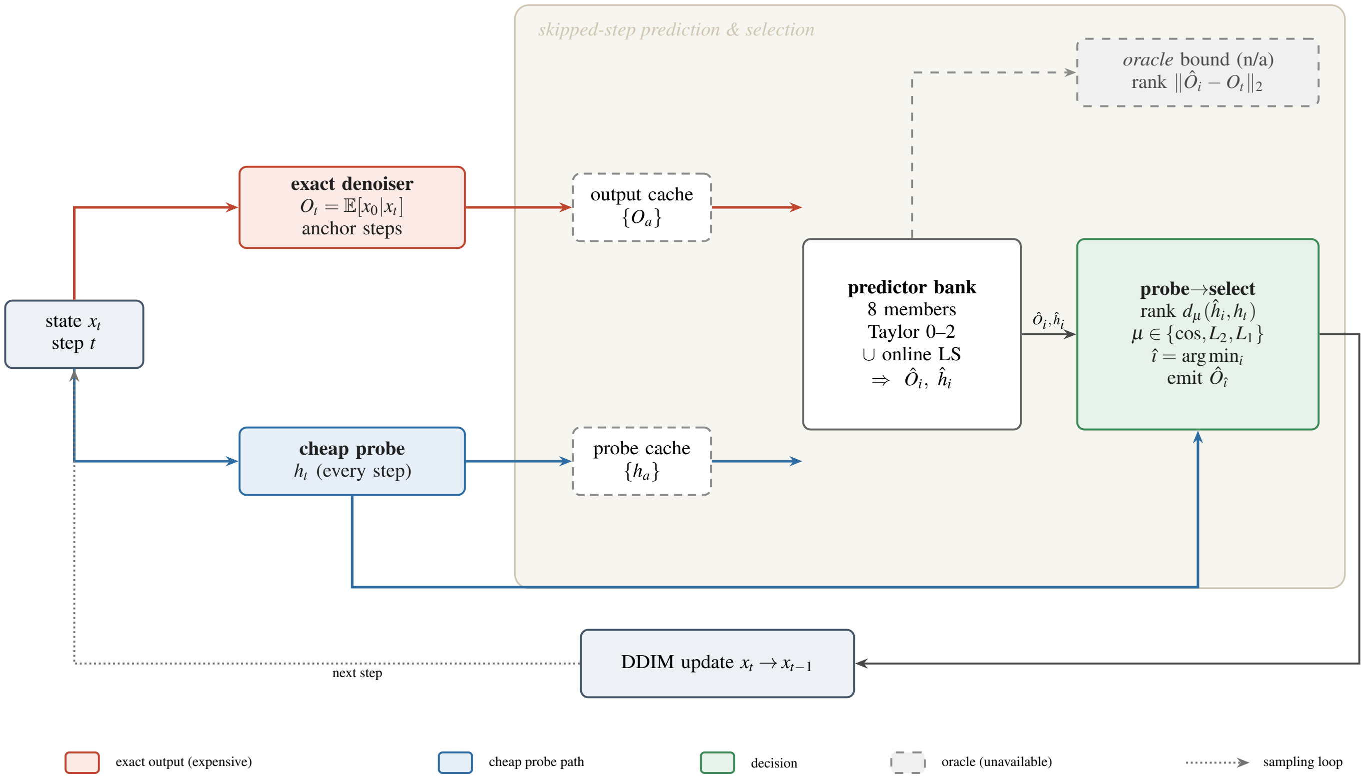


**Figure 1: Architecture of the exact-residual testbed and the probe-then-select mechanism.** Because the prior is a Gaussian mixture, the true denoiser output $O_t$ (red, "expensive") is available in closed form, and a cheap hard-assignment *probe* $h_t$ (blue) plays the role of a first-layer readout. At a skipped step the predictor bank extrapolates the cached outputs and probes; the selector ranks predictors by a proxy distance on the probe and applies the winner's output. The *oracle* (dashed) ranks by the true output error and is never deployable—it bounds how much any proxy could gain.

so we test the *mechanism*, not a particular codebase.

## 3 The exact-residual testbed

We want proxy fidelity to be *measurable*, which on a trained network it is not (computing the true skipped output defeats the purpose). We therefore use a prior for which the exact denoiser is closed-form. Data are drawn from a $K$-component isotropic Gaussian mixture in $\mathbb{R}^d$; under the variance-preserving forward process $x_t = \sqrt{\bar{\alpha}_t}\,x_0 + \sqrt{1-\bar{\alpha}_t}\,\varepsilon$ the posterior mean is exactly

$$O_t \equiv \mathbb{E}[x_0 \mid x_t] = \sum_{k=1}^{K} r_k(x_t,t)\left[\mu_k + c_t\,(x_t - \sqrt{\bar{\alpha}_t}\mu_k)\right], \quad (1)$$

with responsibilities $r_k \propto w_k\,\mathcal{N}\big(x_t; \sqrt{\bar{\alpha}_t}\mu_k, (\bar{\alpha}_t\sigma^2 + 1 - \bar{\alpha}_t)I\big)$ and $c_t = \sqrt{\bar{\alpha}_t}\sigma^2/(\bar{\alpha}_t\sigma^2 + 1 - \bar{\alpha}_t)$. This is the "expensive" output. The cheap *probe* $h_t$ is the hard-assignment denoiser (the $\arg\max_k r_k$ term only): the analog of a shallow first-layer readout that sees an approximate, not exact, decision. Tokens are i.i.d. draws sharing the mixture; DDIM steps use $O$ at anchors and a predicted $\hat{O}$ at skipped steps (Fig. 1).

**Predictors.** A family of 8: fixed finite-difference Taylor extrapolators of order 0–2 at horizons 1–2, plus online least-squares fits (degree 2–3 over windows of 5–6 anchors). **Proxy-then-select.** Each predictor extrapolates the cached outputs ($\hat{O}_i$) and cached probes ($\hat{h}_i$); the selector picks per token $\hat{\imath} = \arg\min_i d_\mu(\hat{h}_i, h_t)$ for proxy $\mu \in \{\cos, L_2, L_1\}$ and applies $\hat{O}_{\hat{\imath}}$. **Oracle** instead minimises the true error $\|\hat{O}_i - O_t\|_2$. **Metrics.** (a) rank correlation between proxy error and true error across the family (does the proxy rank predictors correctly?); (b) energy distance [7] between accelerated and full samplers' final samples (end-to-end distributional fidelity). We use 12 seeds for the core study and 5 per regime cell, with bootstrap 95% CIs throughout.

## 4 The selection problem has no gap

Let $f_n(i)$ be the true reconstruction error of predictor $i$ on token $n$ at a skipped step. TAP minimises $\sum_n f_n(\pi_n)$ over assignments $\pi$. Three properties hold: the objective is *separable* (token $n$'s cost depends only on $\pi_n$), the choices are *equal-cost* (every predictor is one cheap extrapolation; no shared budget), and the feasible set is a *product* set. Hence

$$\min_{\pi} \sum_n f_n(\pi_n) = \sum_n \min_i f_n(i), \quad (2)$$

i.e. greedy per-token $\arg\min$ *is* the global optimum. A regret bound or knapsack reframing over the joint assignment therefore attacks a problem with zero optimality gap—the flaw in our own pilot's headline. The real regret is not combinatorial but *statistical*: we never observe $f_n(i)$; we observe a proxy $g_n(i) = d_\mu(\hat{h}_i, h)$ and deploy $\hat{\pi}_n = \arg\min_i g_n(i)$. The excess error $\mathbb{E}[f_n(\hat{\pi}_n) - \min_i f_n(i)]$ is governed entirely by the rank agreement between $g$ and $f$. The levers that matter are those that raise it: a better metric $\mu$, or a richer probe $h$.

**Why $L_2$ tracks the residual better than cosine.** The true

**Table 1:** Core study (12 seeds, stride-3 base regime). Energy distance to the full sampler (lower is better); paired differences with 95% bootstrap CIs.

| quantity | value | paired CI |
|---|---|---|
| proxy Spearman, cosine | 0.315 | — |
| proxy Spearman, $L_2$ | 0.331 | $+0.016\,(+.014, +.019)$ |
| top-1 pick acc, cosine | 0.393 | — |
| top-1 pick acc, $L_2$ | 0.421 | $+0.028\,(+.024, +.032)$ |
| energy: Taylor family | 0.0210 | — |
| energy: online family | 0.0249 | $-0.004\,(-.006, -.002)$ |
| energy: random select | 0.0281 | — |
| energy: naive best-fixed | 0.0164 | — |
| energy: probe-then-select | 0.0207 | $+0.004\,(+.002, +.006)$ |
| energy: oracle | 0.0112 | $-0.005\,(-.007, -.003)$ |

error is an $L_2$ residual $\|\hat{O}_i - O\|_2$. Cosine scores only the *angle*, discarding magnitude: $d_{\cos} = 1 - \langle \hat{h}_i, h\rangle / (\|\hat{h}_i\|\,\|h\|)$. When predictors differ mainly in how far they over- or under-shoot—the dominant failure mode as the extrapolation horizon grows—the discriminating signal is radial, and cosine is blind to exactly it, whereas $L_2$ (and $L_1$) retain it. If the probe-to-output map is locally affine near the anchors, magnitude ordering is approximately preserved and the $L_2$ proxy is the better-aligned surrogate. This predicts the $L_2$ edge should *grow* with acceleration and noise (more radial error), which the sweep confirms (§6). We state this as mechanism, not theorem: the inequality holds empirically in $21/21$ regimes.

## 5 What survives multi-seed pressure

Table 1 is the core verdict. $L_2$ **beats cosine** in rank fidelity and top-1 pick accuracy, robustly (CI excludes 0) but modestly: a 5% relative gain in Spearman, not the $\sim$4.4$\times$ the single-seed pilot suggested. $L_1$ matches $L_2$, confirming the gain is about *using magnitude*, not cosine specifically. **Online predictors lose**: the online-least-squares family is reliably worse than fixed Taylor, and pooling it with Taylor does not improve over Taylor alone. **Probe-then-select loses to a strong fixed predictor** here ($+0.004$ energy), yet an *oracle* clearly beats naive ($-0.005$): the headroom is real but the probe proxy is too weak to reach it. The bottleneck is proxy fidelity, not the family or the rule.

## 6 When each lever helps: a regime sweep

We sweep acceleration (cache stride 2–8), dimension (2–32), mixture separation, noise scale, and component count $K$—21 regimes, 5 seeds each (Fig. 3). **The $L_2$ edge is universal but small**: positive and significant in $21/21$ regimes, growing from $+0.012$ at mild settings to $+0.034$ at stride 8 and $+0.024$ at high noise, consistent with the radial-error mechanism. **Online never wins**: Taylor is significantly better in $19/21$ regimes and online in $0/21$; the best single fixed predictor is a Taylor 100% of the time. **Probe adaptivity is an aggressive-acceleration phenomenon**: probe-then-select beats naive best-fixed in exactly one regime, stride 8, where fixed extrapolation breaks down enough that per-token switching pays (probe$-$naive $= -0.074$). Everywhere milder the two curves lie on top of each other (Fig. 3a), so the measured gains of "adaptive" caching at moderate speedups come from the *speedup*, not the adaptivity. Selection headroom grows with acceleration (Fig. 3c), so aggressive regimes are where a better proxy would matter most. **Proxy fidelity is the real lever**: replacing the shallow hard probe with a richer top-*m* soft readout raises rank fidelity monotonically—$0.33 \to 0.68 \to 0.73 \to 1.0$ (Fig. 3d). A real DiT's learned first-layer probe sits far up this curve (our companion study measures Spearman $\approx$0.77 on a trained DiT [14]), so the testbed *lower-bounds* the probe's value and our $L_2$ gap is a conservative estimate.

## 7 A 30-test adversarial suite

We pre-registered 30 tests, each trying to break a claim: testbed-correctness asserts (exact denoiser limits, energy(full, full)=0, no leakage of the true error into selection), the multi-seed core comparisons, the 21-regime crossovers, acceleration-Pareto monotonicity, a budget/abstaining selector's compute–fidelity monotonicity, and the probe-richness curve. 26 **pass,** 0 **fail,** 4 **are honest notes**: the $L_2$ edge is small not $\sim$4$\times$; the absolute proxy Spearman is only moderate; the end-to-end quality gain of $L_2$ over cosine at the selection level is negligible; and at mild acceleration the speedup, not adaptivity, drives measured gains. No test designed to break a surviving claim succeeded, and the two retired levers failed exactly the tests meant to catch them.

## 8 Discussion

The contribution is subtractive. Of three plausible improvements to a token-adaptive predictor, one (online predictors) is a mirage, one (probe adaptivity) is real only in an aggressive-acceleration corner, and one ($L_2$ proxy) is a small but dependable gain best read as "stop discarding magnitude." The reframing that makes these cohere is that TAP's selection is already optimal given true errors, so the whole game is proxy fidelity—set by the metric and, far more, by probe richness. This turns a method-tuning question into a measurement question, and the exact-residual testbed is what makes the measurement possible.

## 9 Limitations

The denoiser is an analytic Gaussian mixture, not a trained network, and the probe is a shallow hard-assignment readout; both are chosen so the truth is computable, but both make the probe a *weaker* ranker than a real DiT's learned first layer (Fig. 3d and [14]). Absolute energy-distance magnitudes are testbed specific; we rely on paired, within-regime comparisons only. TAP has no public code, so the mechanism, not a reference implementation, is what we test. Whether the small $L_2$ edge and the aggressive-acceleration probe advantage grow or shrink on large latent DiTs is open and needs a GPU cluster we did not use.

## 10 Conclusion

We tried to improve a training-free diffusion predictor and mostly falsified our own ideas, which is the point: under multi-seed, multi-regime pressure, online predictors do not beat Taylor, probe adaptivity pays only at aggressive acceleration, and an $L_2$ proxy gives a small, robust edge over cosine because it stops discarding the magnitude that the true residual depends

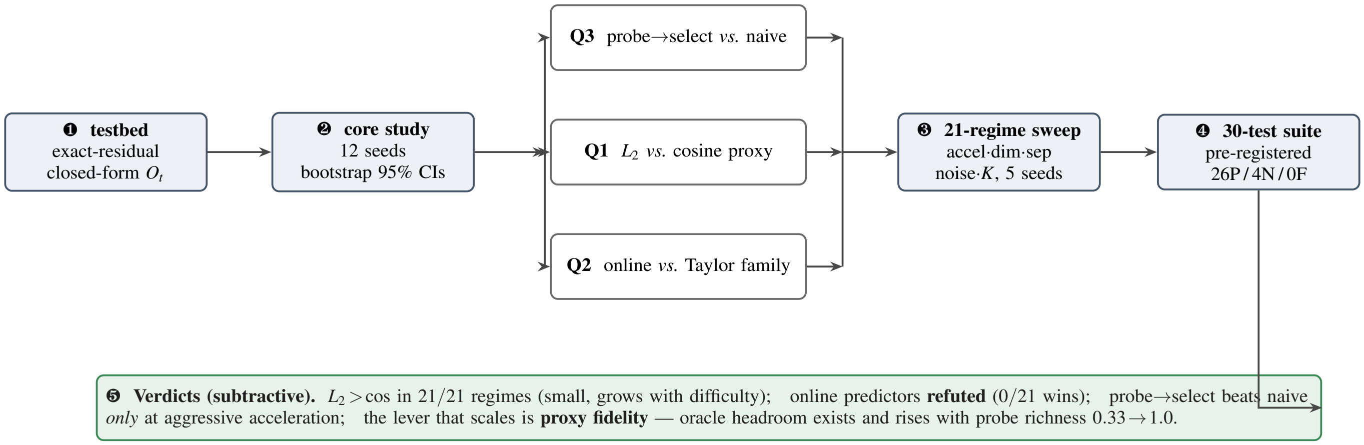


**Figure 2: Experimental workflow.** The exact testbed feeds a 12-seed core study that isolates three questions (Q1–Q3); each is then stress-tested across a 21-regime sweep and a pre-registered 30-test adversarial suite. Every comparison is paired and bootstrapped. The surviving conclusion is subtractive: only the $L_2$ proxy holds (small), and the real lever is proxy fidelity, set by probe richness.

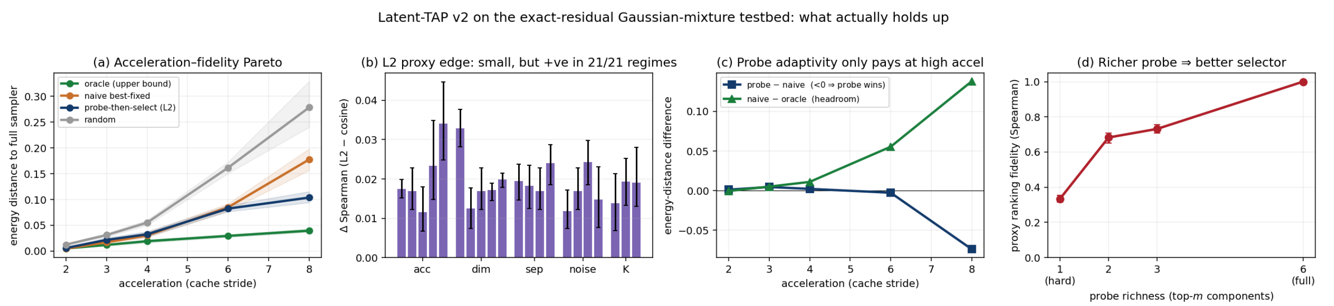


**Figure 3: (a)** Acceleration–fidelity Pareto: probe-then-select tracks the naive best-fixed predictor until high stride, where it finally separates toward the oracle. **(b)** The $L_2$ proxy edge over cosine is small but positive in all 21 regimes, widening with acceleration and noise. **(c)** Probe adaptivity only pays at aggressive acceleration (probe−naive crosses below 0 at stride 8), exactly where selection headroom (naive−oracle) is largest. **(d)** Proxy fidelity rises monotonically with probe richness, from a shallow hard probe (0.33) to the full output (1.0)—the dimension along which a real learned DiT probe improves.

on. Because per-token selection is provably already optimal, the only lever that scales is proxy fidelity—set by the metric and, decisively, by probe richness. Evaluate these methods on trained networks with fidelity-to-base-model metrics, and treat the probe, not the selection rule, as the thing to improve.

**Reproducibility.** All code and results run locally in pure NumPy: `latap_v2.py` (testbed), `run_core.py`/`agg_core.py` (12-seed core), `sweep.py`/`sweep_agg.py` (21-regime sweep), `probe_depth.py` (richness), `pressure_v2.py` (30-test suite), `plot_v2.py`; data in `results/`. See the repository README for exact commands.

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